

# **PHYSICS**

## **Written examination 2**

### **FORMULA SHEET**

#### **Directions to students**

Detach this formula sheet before commencing the examination.

This formula sheet is provided for your reference.

|    |                                                |                                                                                                                 |
|----|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| 1  | photoelectric effect                           | $E_{K\max} = hf - W$                                                                                            |
| 2  | photon energy                                  | $E = hf$                                                                                                        |
| 3  | photon momentum                                | $p = \frac{h}{\lambda}$                                                                                         |
| 4  | de Broglie wavelength                          | $\lambda = \frac{h}{p}$                                                                                         |
| 5  | resistors in series                            | $R_T = R_1 + R_2$                                                                                               |
| 6  | resistors in parallel                          | $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$                                                                 |
| 7  | magnetic force                                 | $F = I l B$                                                                                                     |
| 8  | electromagnetic induction                      | emf : $\varepsilon = -N \frac{\Delta\Phi}{\Delta t}$ flux: $\Phi = BA$                                          |
| 9  | transformer action                             | $\frac{V_1}{V_2} = \frac{N_1}{N_2}$                                                                             |
| 10 | AC voltage and current                         | $V_{\text{RMS}} = \frac{1}{\sqrt{2}} V_{\text{peak}} \quad I_{\text{RMS}} = \frac{1}{\sqrt{2}} I_{\text{peak}}$ |
| 11 | voltage; power                                 | $V = RI \quad P = VI$                                                                                           |
| 12 | transmission losses                            | $V_{\text{drop}} = I_{\text{line}} R_{\text{line}} \quad P_{\text{loss}} = I_{\text{line}}^2 R_{\text{line}}$   |
| 13 | mass of the electron                           | $m_e = 9.1 \times 10^{-31} \text{ kg}$                                                                          |
| 14 | charge on the electron                         | $e = -1.6 \times 10^{-19} \text{ C}$                                                                            |
| 15 | Planck's constant                              | $h = 6.63 \times 10^{-34} \text{ J s}$<br>$h = 4.14 \times 10^{-15} \text{ eV s}$                               |
| 16 | speed of light                                 | $c = 3.0 \times 10^8 \text{ m s}^{-1}$                                                                          |
| 17 | Acceleration due to gravity at Earth's surface | $g = 10 \text{ m s}^{-2}$                                                                                       |

### Detailed study 3.1 – Synchrotron and applications

|    |                                                                    |                             |
|----|--------------------------------------------------------------------|-----------------------------|
| 18 | energy transformations for electrons in an electron gun (<100 keV) | $\frac{1}{2} m v^2 = eV$    |
| 19 | radius of electron beam                                            | $r = m v / eB$              |
| 20 | force on an electron                                               | $F = evB$                   |
| 21 | Bragg's law                                                        | $n\lambda = 2d \sin \theta$ |
| 22 | electric field between charged plates                              | $E = \frac{V}{d}$           |

**Detailed study 3.2 – Photonics**

|    |                 |                           |
|----|-----------------|---------------------------|
| 23 | band gap energy | $E = \frac{hc}{\lambda}$  |
| 24 | Snell's law     | $n_1 \sin i = n_2 \sin r$ |

**Detailed study 3.3 – Sound**

|    |                                 |                                                                                                                                                                |
|----|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 25 | speed, frequency and wavelength | $v = f\lambda$                                                                                                                                                 |
| 26 | intensity and levels            | <p>sound intensity level</p> $(\text{in dB}) = 10 \log_{10} \left( \frac{I}{I_0} \right)$ <p>where <math>I_0 = 1.0 \times 10^{-12} \text{ W m}^{-2}</math></p> |

**Prefixes/Units**p = pico =  $10^{-12}$ n = nano =  $10^{-9}$  $\mu$  = micro =  $10^{-6}$ m = milli =  $10^{-3}$ k = kilo =  $10^3$ M = mega =  $10^6$ G = giga =  $10^9$ t = tonne =  $10^3 \text{ kg}$ **END OF DATA SHEET**